8/4/2026

Recall: gamma fn

$$\Gamma(k) = \begin{cases} \int_0^\infty e^{-x} x^{k-1}\,dx \\ 2\int_0^\infty e^{-x^2} x^{2k-1}\,dx \end{cases}$$

$k > 0$

1. $\Gamma(1) = 1$ ✓
2. $\Gamma(n+1) = n\,\Gamma(n)$ ✓
3. $\Gamma(n) = (n-1)!$

if n is a positive integer

Proof: We have

$$\Gamma(n) = (n-1)\,\Gamma(n-1)$$

[$\because$ $\Gamma(n+1) = n\,\Gamma(n)$, property 2]

ie $\Gamma(n) = (n-1)\left[(n-2)\,\Gamma(n-2)\right]$

$= (n-1)(n-2)(n-3)(n-4)\cdots 1\cdot\Gamma(1)$

$= (n-1)(n-2)(n-3)(n-4)\cdots 1.1$ ($\because \Gamma(1)=1$)

$\Rightarrow \Gamma(n) = (n-1)!$

Eg: $\Gamma(3) = (3-1)! = 2!$

$\Gamma(7) = 6!$

$\Gamma(7/2) = ?$

4. $\Gamma(p)\,\Gamma(1-p) = \dfrac{\pi}{\sin \pi p}$, $\underline{0 < p < 1}$

$\boxed{\text{eg1}}$ $p = \frac{2}{3}$

$1 - p = \frac{1}{3}$

$$\therefore \Gamma\left(\tfrac{2}{3}\right)\cdot\Gamma\left(\tfrac{1}{3}\right) = \frac{\pi}{\sin \pi \frac{2}{3}}$$

$$= \frac{\pi}{\sin \frac{2\pi}{3}}$$

$$= \frac{\pi}{\sqrt{3}} \times 2 \;//$$

<u>Eg2</u>: $p = \frac{1}{2}$

<u>eg2</u>: $p = \frac{1}{2}$

$$\Gamma(p)\,\Gamma(1-p) = \frac{\pi}{\sin \pi p}$$

ie $\Gamma\left(\frac{1}{2}\right)\Gamma\left(\frac{1}{2}\right) = \dfrac{\pi}{\sin \frac{\pi}{2}}$

$$\Rightarrow \left[\Gamma\left(\tfrac{1}{2}\right)\right]^2 = \frac{\pi}{1}$$

$$\Rightarrow \underline{\Gamma\left(\tfrac{1}{2}\right) = \sqrt{\pi}}$$

Find $\Gamma\left(\frac{3}{2}\right) = \left(\frac{3}{2} - 1\right)\Gamma\left(\frac{3}{2} - 1\right)$

$$= \frac{1}{2}\Gamma\left(\tfrac{1}{2}\right)$$

$$= \frac{1}{2}\sqrt{\pi} \;//$$

$$\cdot\ \Gamma\left(\frac{7}{2}\right) = \left(\frac{7}{2}-1\right)\Gamma\left(\frac{7}{2}-1\right)$$

$$= \frac{5}{2}\Gamma\left(\frac{5}{2}\right)$$

[$\Gamma(n) = (n-1)\Gamma(n-1)$ property]

$$= \frac{5}{2}\left(\frac{5}{2}-1\right)\Gamma\left(\frac{5}{2}-1\right)$$

$$= \frac{5}{2}\cdot\frac{3}{2}\cdot\Gamma\left(\frac{3}{2}\right)$$

$$= \frac{5}{2}\cdot\frac{3}{2}\cdot\frac{1}{2}\Gamma\left(\frac{1}{2}\right)$$

$$= \frac{5\cdot 3\cdot 1}{2^3}\sqrt{\pi} \longrightarrow \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$

$$\Rightarrow \Gamma\left(\frac{7}{2}\right) = \frac{15}{8}\sqrt{\pi}$$

$$\cdot\ \Gamma\left(\frac{11}{2}\right) = \frac{9}{2}\cdot\frac{7}{2}\cdot\frac{5}{2}\cdot\frac{3}{2}\cdot\frac{1}{2}\Gamma\left(\frac{1}{2}\right)$$

$$= \frac{945}{32}\sqrt{\pi}$$

$$\boxed{\Gamma(n) \neq \sqrt{\Gamma(n^2)}}$$

$$\cdot\ \Gamma\left(\frac{5}{4}\right) = \frac{1}{4}\Gamma\left(\frac{1}{4}\right)$$

$\therefore$

Examples

1. Find $\int_0^\infty e^{-x^5}\,dx$

Sol^n: $\left[\ e^{-t} \rightarrow e^{-x^5}\ \right]$ (integral)

put $x^5 = t$

$\Rightarrow x = t^{1/5}$

$\Rightarrow dx = \frac{1}{5} t^{-4/5}\,dt$

x	0	∞
$t = x^5$	0	∞

$$\therefore \int_0^\infty e^{-x^5}\,dx = \int_0^\infty e^{-t} \cdot \frac{1}{5} t^{-4/5}\,dt$$

$$= \frac{1}{5} \int_0^\infty e^{-t}\, t^{-4/5}\,dt$$

$$\left[\ \Gamma(k) = \int_0^\infty e^{-t}\, t^{k-1}\,dt\ \right]$$

$$\therefore \int_0^\infty e^{-x^5}\,dx = \frac{1}{5}\Gamma\left(\frac{1}{5}\right)$$

$$\left[\begin{aligned} k-1 &= -4/5 \\ \Rightarrow k &= -\frac{4}{5}+1 \\ &= 1/5 \end{aligned}\right]$$

2. Ev. $\int_0^\infty e^{-ax^n}\,dx$

3. Ev. $\int_0^\infty e^{-h^2x^2}\,dx$

4. HW Ev. $\int_0^\infty e^{-x^2/4}\,dx$

[Ans: $\sqrt{\pi}$]

5. Ev. $\int_0^\infty x^m e^{-ax^n}\,dx$

Ans: $\frac{1}{n}\left(\frac{1}{a^{\frac{m+1}{n}}}\right)\Gamma\left(\frac{m+1}{n}\right)$

$ax^n = t$

$x = \left(\frac{t}{a}\right)^{1/n}$

$dx = \frac{1}{a^{1/n}}\frac{1}{n}\cdot t^{\frac{1}{n}-1}$

2. Sol^n: put $ax^n = t$

$\Rightarrow x^n = \frac{t}{a}$

$\Rightarrow x = \left(\frac{t}{a}\right)^{1/n}$

$\Rightarrow dx = \frac{1}{n a^{1/n}} t^{\frac{1}{n}-1}\,dt$

x	0	∞
$t = ax^n$	0	∞

$-2 \quad -1 \quad 0$

$\therefore \int_0^\infty e^{-ax^n}\,dx = \int_0^\infty e^{-t}\,\frac{1}{n a^{1/n}}\left(\left(\frac{t}{a}\right)^{1/n}\right)^m \cdot t^{\frac{1}{n}-1}\,dt$

$= \frac{1}{n a^{1/n + \frac{m}{n}}}\int_0^\infty e^{-t}\, t^{\left(\frac{m}{n}+\frac{1}{n}\right)-1}\,dt$

$\Rightarrow \int_0^\infty e^{-ax^n}\,dx = \frac{1}{n a^{1/n}}\Gamma\left(\frac{1}{n}\right)$

3. put $h^2x^2 = t$

$\Rightarrow x^2 = \frac{t}{h^2}$

$\Rightarrow x = \left(\frac{t}{h^2}\right)^{1/2} = \frac{1}{h}t^{1/2}$

$t^{-1/2}\,dt$

$dx = \int_0^\infty e^{-t}\cdot\frac{1}{2h}t^{1/2}\,dt$

$= \frac{1}{2h}\Gamma\left(\frac{1}{2}\right) = \frac{1}{2h}\sqrt{\pi}$